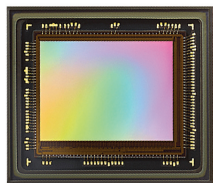


erating low latency results with higher accuracy. This device can enable engineers to build smaller and cost-effective products without compromising on computing power. Compared to other GPUs the Kinara processor offers better performance/watt, making it suitable for a wide variety of applications, such as in smart retail, smart cities, automotive, and Industry 4.0.

Kinara
<https://kinara.ai>

Sensor for machine vision applications

Prophesee's third-generation Metavision sensor is suitable for machine vision applications. It offers a throughput of more than 1000 objects/second with a power consumption of 3nW/s.



The sensor has a capturing rate equivalent to 10k frames per second and the ability to capture images in a dark environ-

ment. These features make the sensor suitable for use in applications such as ultra-high-speed counting, vibration measurement, and kinematic monitoring. Each individual pixel in a Metavision sensor embeds its own intelligence processing, enabling them to activate themselves independently for triggering events. The sensor comes in a package measuring just 13mm x15mm and captures an image with a resolution of 640x480 pixels. It can enable designers to build a highly efficient machine vision product.

Prophesee
<https://www.prophesee.ai>

High-efficiency synchronous buck converter

SZPL3002A from Silanna Semiconductor is the world's first buck converter IC with a built-in USB PD/FC port controller to feature intelligent, adaptive power sharing. The IC comes in a compact QFN package measuring just 5mm x 5mm. The SZPL3002A has an efficiency of over 98% and offers one fully programmable contract set

plus four pre-programmed sets that are chosen with an external resistor value. The IC also features protection features,



such as power throttling at elevated temperatures. The high-efficiency synchronous buck converter can reduce component count in the circuit and simplifies power-sharing design in multi-port fast chargers.

Silanna Semiconductor
<https://silanna.com>

SoC for ultra-low-power IoT devices

SiWx917 system-on-chip (SoC) from Silicon Labs is designed to create ultra-low-power IoT devices. The SoC can deliver more compute, enable faster AI/ML, and have higher security in a single compact package.



The SiWx917 is powered by an ARM Cortex M4 processor with FPU application MCU up to 180MHz. The

SoC is suitable for designing low-power IoT wireless devices using Wi-Fi 6 plus, Bluetooth Low Energy (LE) 5.1, Matter, and IP networking for secure cloud connectivity. The SoC can help developers simplify their designs by reducing development costs, and size, thus reducing the time to market. It supports OFDMA, MU-MIMO, BSS colouring, etc, which allows for faster, more stable network coverage even in crowded areas. The SoC can also be used for applications such as battery-powered locks, asset trackers, and health and fitness monitors.

Silicon Labs
<https://www.silabs.com>

Wi-Fi SoC for ultra-low-power IoT applications

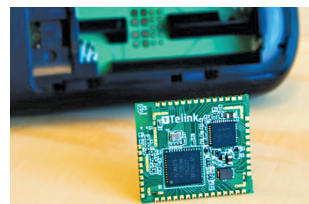
The UBI206 chip can bring Wi-Fi connectivity to your IoT solution while consuming battery equivalent to a BLE chip. Ubilite's UBI206 platform is based on the UBI206 system-on-chip (SoC). It is a highly integrated product that integrates communication, power manage-

ment unit, memory, and processor in a single module. It can enable engineers to create products with Wi-Fi connectivity while having higher battery life than other similar products. The UBI206 platform reduces the BoM cost and simplifies designing an IoT system, therefore reducing the time to market. The UBI206 platform is suitable for ultra-low-power IoT markets, including healthcare, wearables, trackers, and industrial systems monitoring and control.

Ubilite
<https://www.ubilite.com>

Batteryless module with multiprotocol connectivity

TLRS8273-M-EH from Telink is a multiprotocol connectivity module with an in-built energy harvesting capability. The innovative component integrates all the essential components that are required to design a batteryless remote.



It is based on a 48MHz RISC microcontroller that supports Bluetooth 5.1 LE, 802.15.4 (Zigbee/Rf4CE/6LoWPAN/Thread) and 2.4GHz proprietary protocols. With its combined wireless connectivity and energy harvesting on a compact footprint, it makes batteryless operations possible for a wide range of IoT applications. It also offers maximum power point tracking, which can enhance the harvesting capability of the module. The module has 32 GPIOs and features SPI, I2C, USB 2.0, Swire, etc for communication. It also has six channels of differential PWM and an IR transmitter with DMA. It can be used in various IoT applications, including wearables, electronic shelf labels, and remotes.

Telink
<https://www.telink-semi.com>